UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

EEL4712 Digital Design

Lecture 34 – Timing Constraints in FPGA Designs

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Agenda

1. Motivation
2. Timing Constraints
   - Setup
   - Hold time
   - Clock skew
3. Static Timing Analysis
4. Timing Constraints in Prime Time
5. Example of Timing Analysis and Optimization using Prime Time with Quartus

Motivation

- Why Timing Constraints Matter
  - Signals take time to propagate through logic
  - Registers must capture data at the correct clock edge
  - Violating timing → incorrect or unpredictable behavior
- Your design will only behave as your RTL if timing is satisfied!

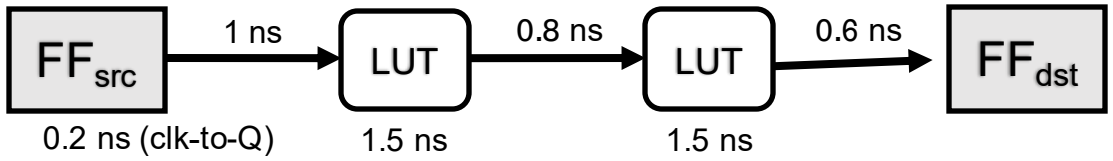
Timing Constraints define how fast your design can operate!

Review: Setup Timing

- Setup requirement:

$$t_{clk_q} + t_{comb} \leq T_{clk} - t_{setup}$$

- Maximum delay constraint
- Ensures data arrives before next clock edge
- Setup Violation → Data arrives too late



t_{setup} = 0 in this example

T_C = 0.2 ns + 1.5 ns + 1.5 ns = 3.2 ns

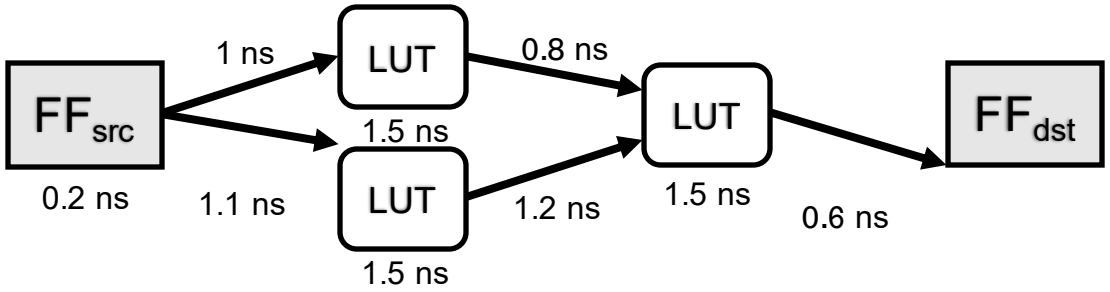
T_{IC} = 1 ns + 0.8 ns + 0.6 ns = 2.4 ns

$T_{FF\text{-}to\text{-}FF} = T_C + T_{IC}$ = 3.2 ns + 2.4 ns = 5.6 ns

frequency = 1 / $T_{FF\text{-}to\text{-}FF}$ = 1 / 5.6 ns = 178 MHz

Setup Timing

- There are frequently multiple paths between same FFs
- Must consider the path with the maximum delay
- Interconnect delays often vary across paths due to routing differences



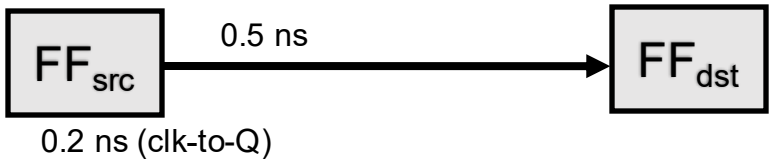
Can you find the critical path?

Review: Hold Timing

- Hold requirement:

$$t_{clk_q} + t_{comb} \geq t_{hold}$$

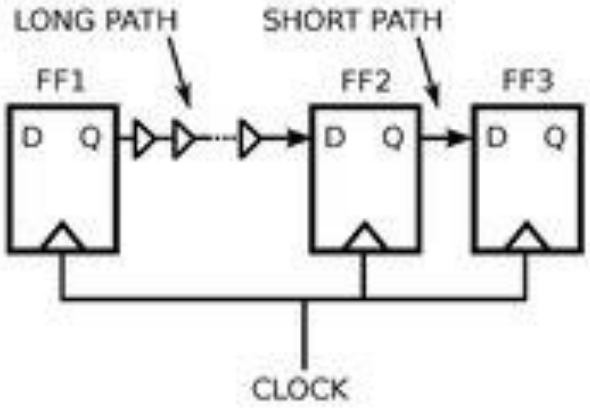
- Minimum delay constraint
- Ensures data does not change too early
- Hold Violation → Data Changes too Early

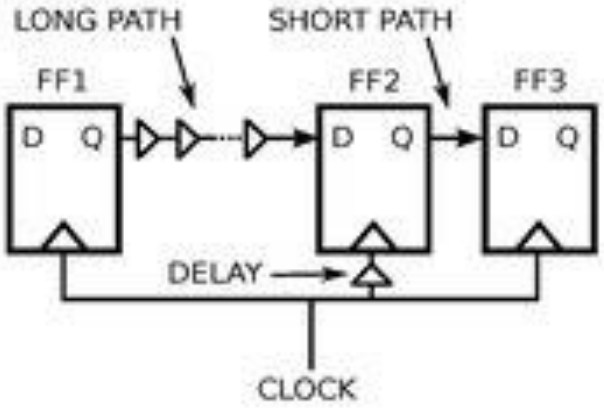


t_{hold} = 1 ns
0.2 ns + 0.5 ns >= 1 ns

Clock Skew

- The clock signal does not arrive to each FF exactly at the same time
- Useful skew can be used to reallocate timing margin between different pairs of FFs



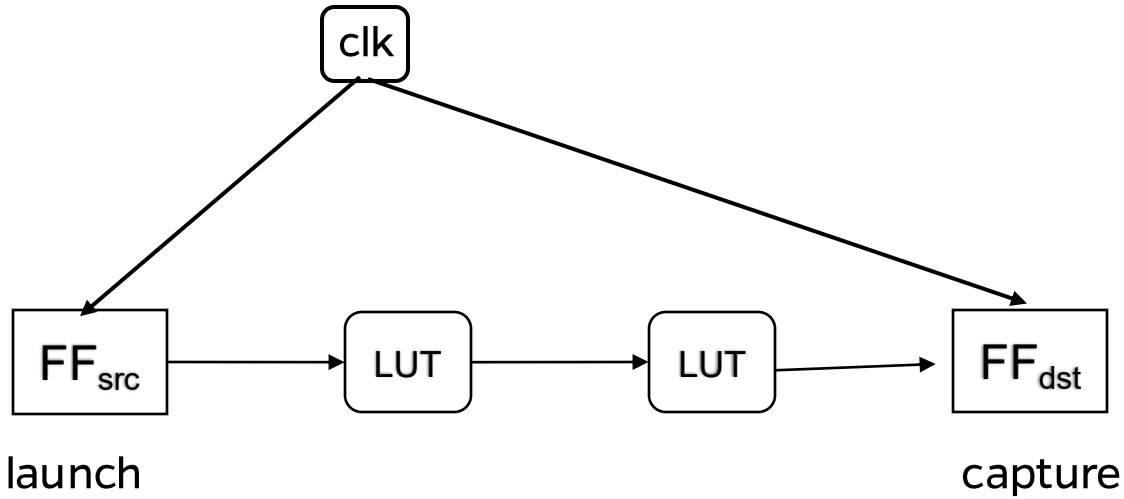


Slack

- Slack = Required Time − Arrival Time
  - Positive slack → Safe
  - Zero slack → Barely meets timing
  - Negative slack → Timing violation
- Critical path = Path with smallest slack
- Timing Quality Measures
  - Worst Negative Slack (WNS)
  - Total Negative Slack (TNS)

Arrival Time and Required Time

- Arrival Time (AT):
  - $AT = T_{clk \rightarrow Q} + T_{logic} + T_{route} + T_{clk_launch}$
- Required Time (RT):
  - $RT = T_{clk} + T_{clk_capture} - T_{setup}$
- Slack:
  - $Slack = RT - AT$



What Is Static Timing Analysis (STA)?

- STA:
  - Does NOT simulate
  - Analyzes all register-to-register paths
  - Evaluates worst-case timing
- Checks:
  - Setup slack
  - Hold slack
  - Critical path

What Do You as a Designer Do?

- Specify the timing constraints
  - Synopsys Design Constraint File (SDC)
- Compile and Synthesize
  - If there are timing violations, modify the design to make it meet the timing constraints
  - If the timing is clean, map the design to board!
- Common Constraints (on next slides)
  - Create clocks
  - Input/output timing
  - False paths
  - Asynchronous clocks

Most Important Constraint: create_clock

- Defines clock period
- 10 ns → 100 MHz
- Without this:
- Tool guesses timing
- Optimization is meaningless

```
create_clock -name clk -period 10 [get_ports clk]
```

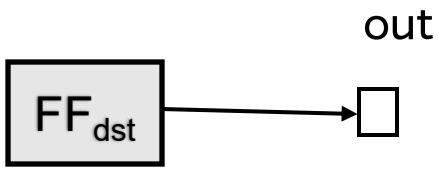
Input Timing Constraints

- External devices matter.
- Example:
  - `set_input_delay -clock clk 2.0 [get_ports data_in]`
- Meaning:
  - Data arrives 2 ns after clock edge
- Without this:
  - FPGA assumes data arrives instantly



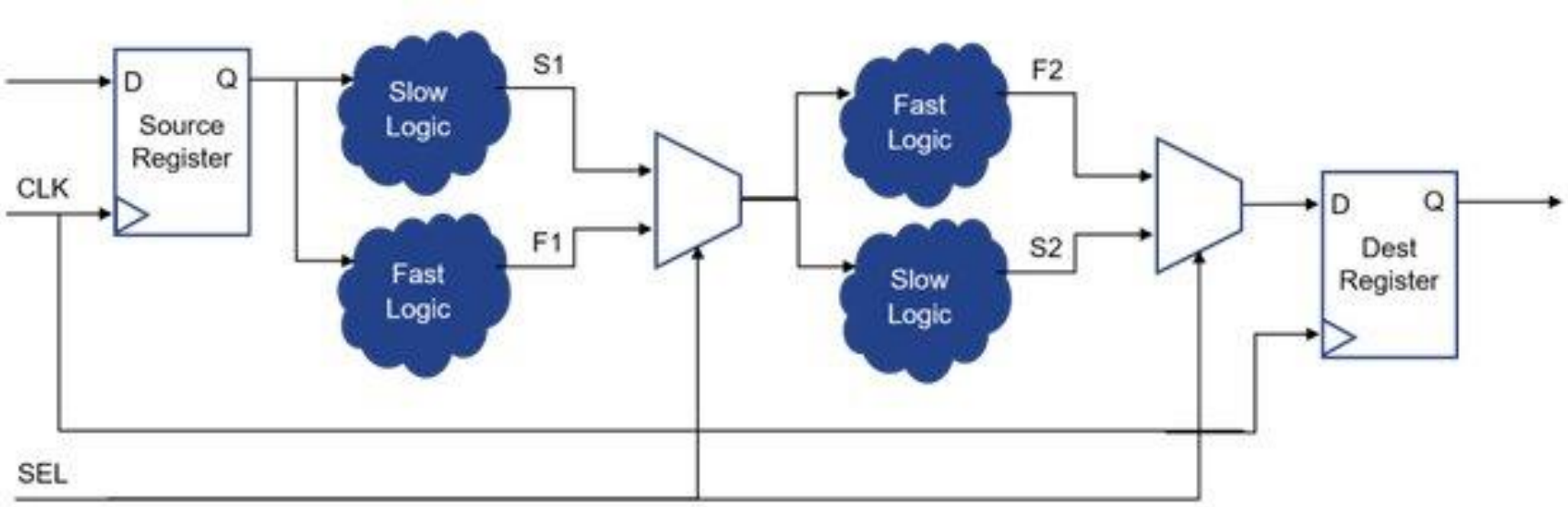
Output Timing Constraints

- Example:
  - `set_output_delay -clock clk 3.0 [get_ports data_out]`
- Meaning:
  - External device requires data 3 ns before its clock edge
- Important for:
  - ADCs
  - DACs
  - SRAM
  - VGA



False Paths

- A false path (FP) occurs when there is a traceable path through a design that is never enabled.
- FP are not required to meet timing constraints



False Paths

- Used for paths that are:
  - Functionally irrelevant
  - Asynchronous resets
  - Debug logic
- Example:
  - `set_false_path -from A -to B`

Overusing false paths hides real bugs.

Multicycle Paths

- Used when logic intentionally takes multiple cycles.
- Example:
  - `set_multicycle_path 2`
- Meaning:
  - Data has 2 clock cycles to arrive
- Common in:
  - Iterative multipliers
  - FSM-based datapaths

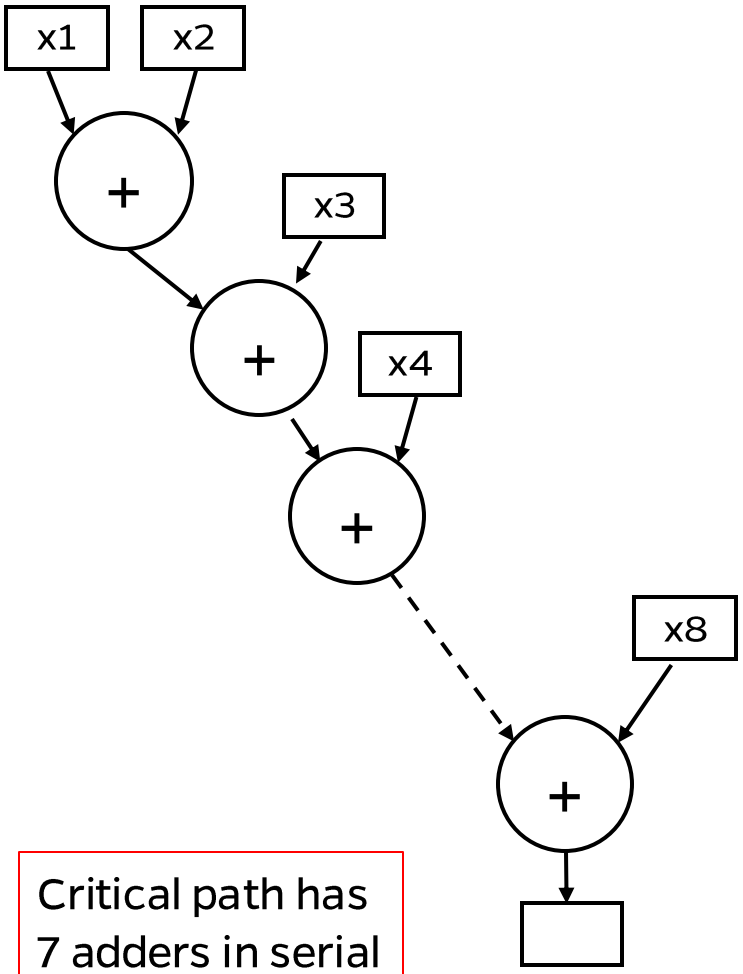
Clock Domain Crossing Constraints

- Timing tools assume clocks are related unless told otherwise.
- For asynchronous clocks:
  - `set_clock_groups -asynchronous`
  -
- Without this:
  - Tool tries to time impossible paths
  - False violations appear

Common Design Mistakes

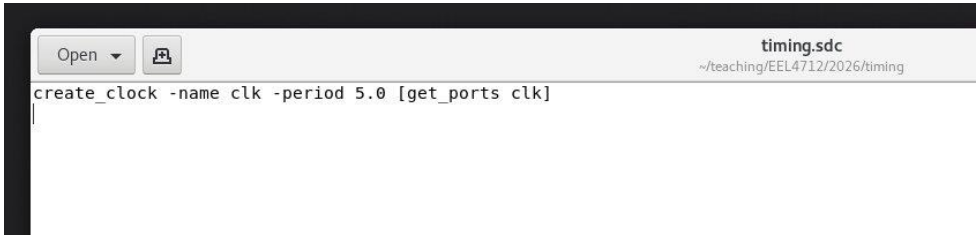
- No clock constraint
- Wrong clock period
- Forgetting I/O delays
- Using false paths to hide violations
- Ignoring hold violations

Walk Through of Timing Example



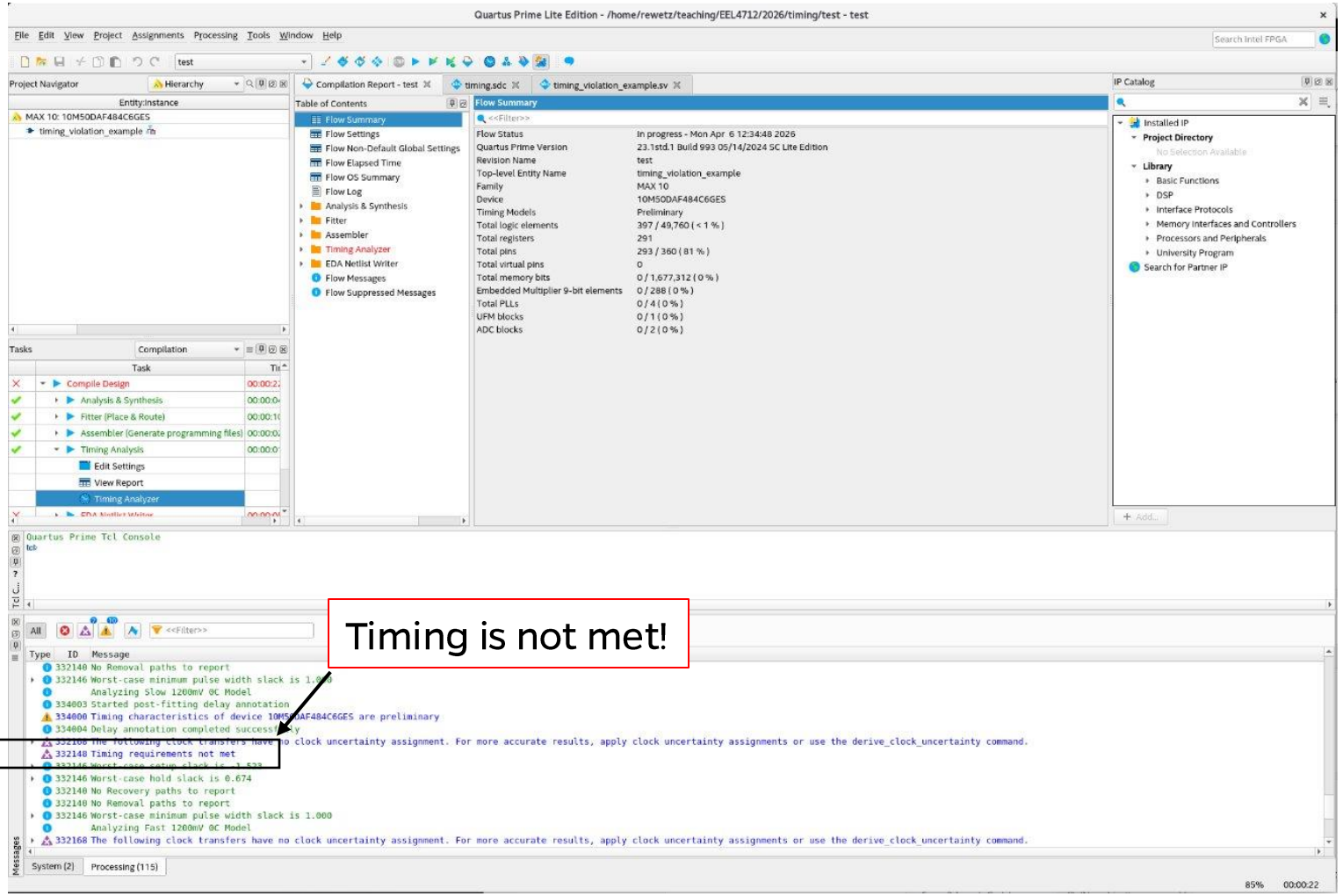
Critical path has
7 adders in serial

Create Design in Quartus
-Add serial adder as top-level
-Add SDC file



Clock period of 5 ns => Fmax 200 MHz

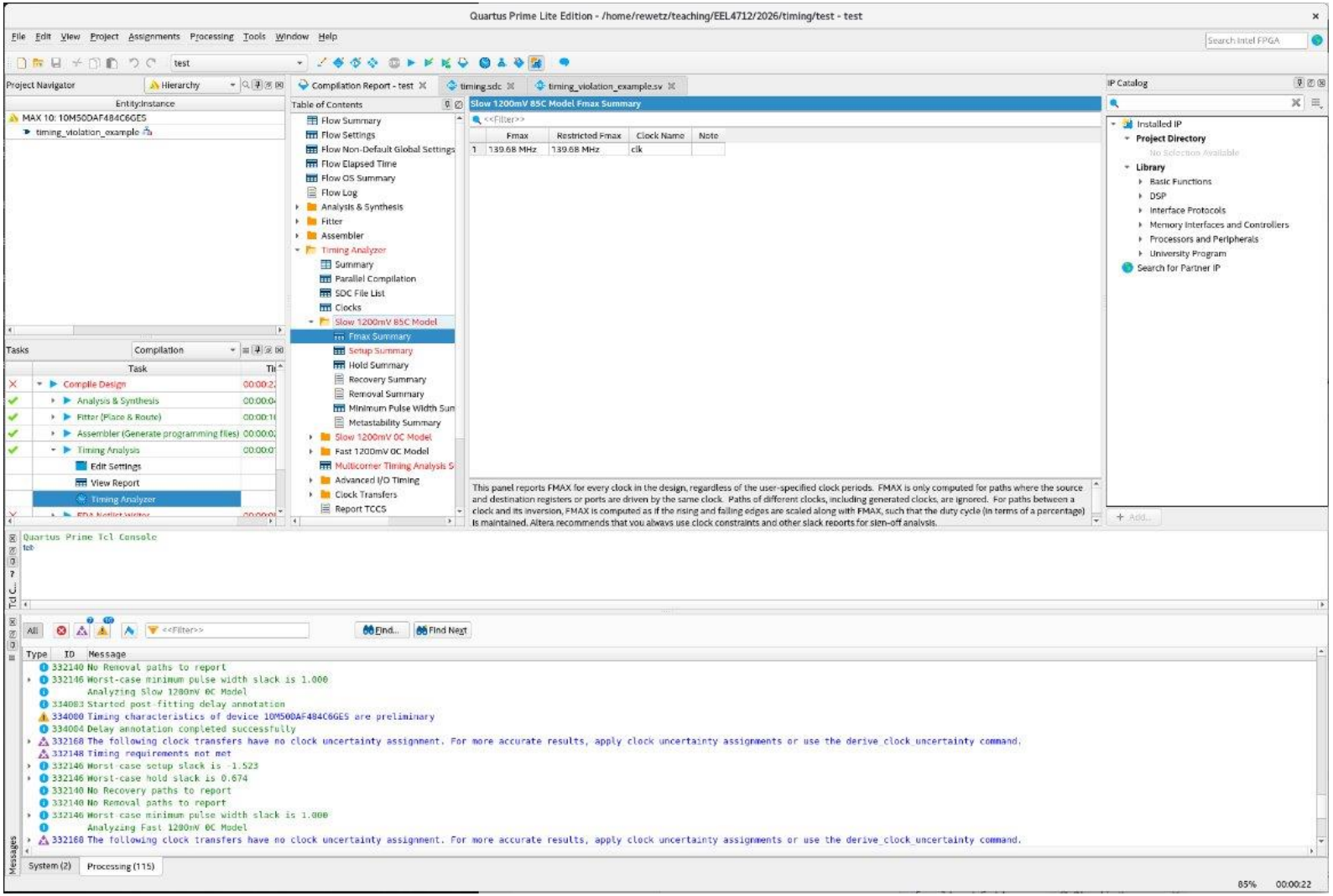
Compiling the Design



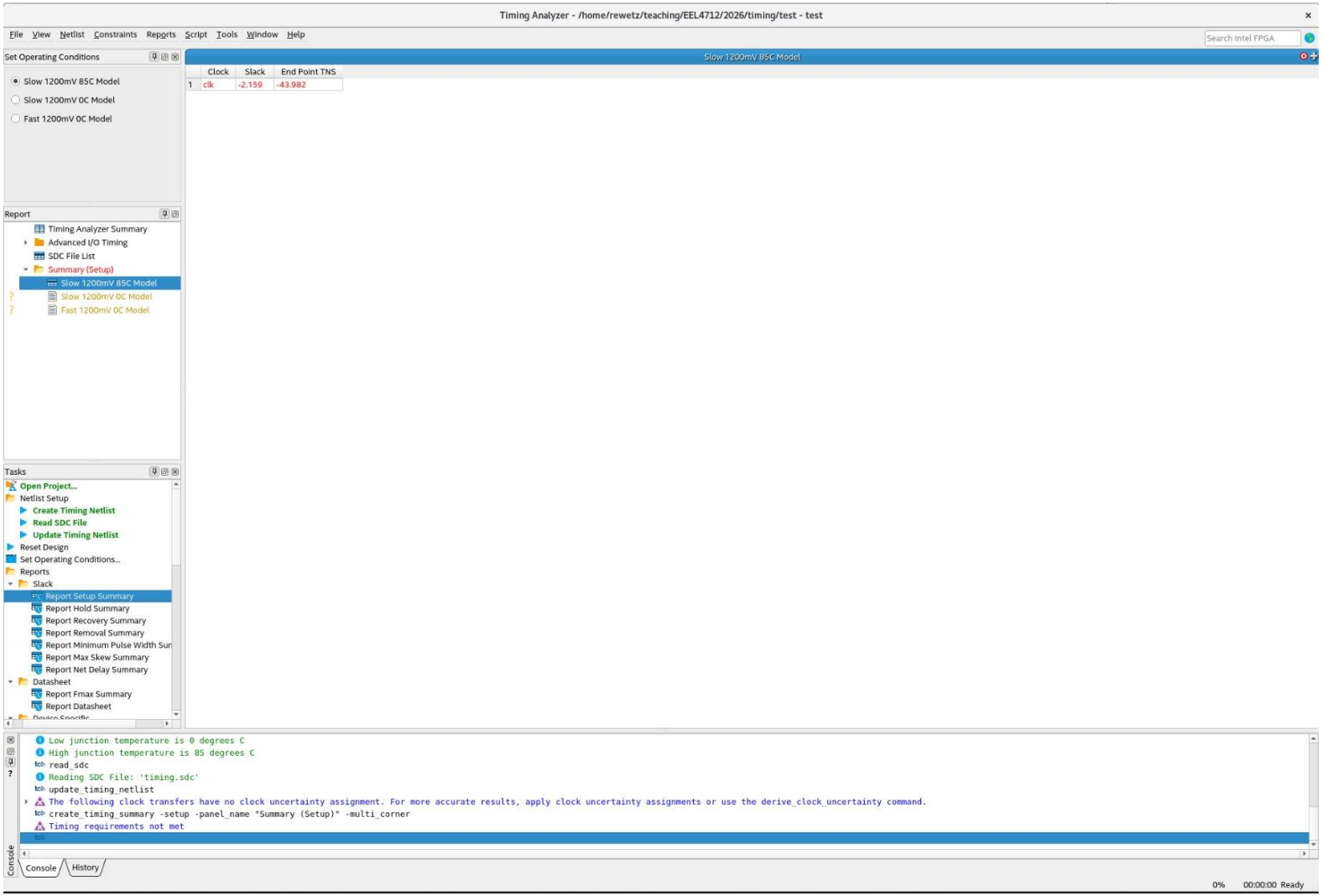
Reading a Timing Report

- Metrics and terms:
  - Worst slack
  - Critical path
  - Launch register
  - Capture register
- Total delay breakdown
  - Logic delay
  - Routing delay

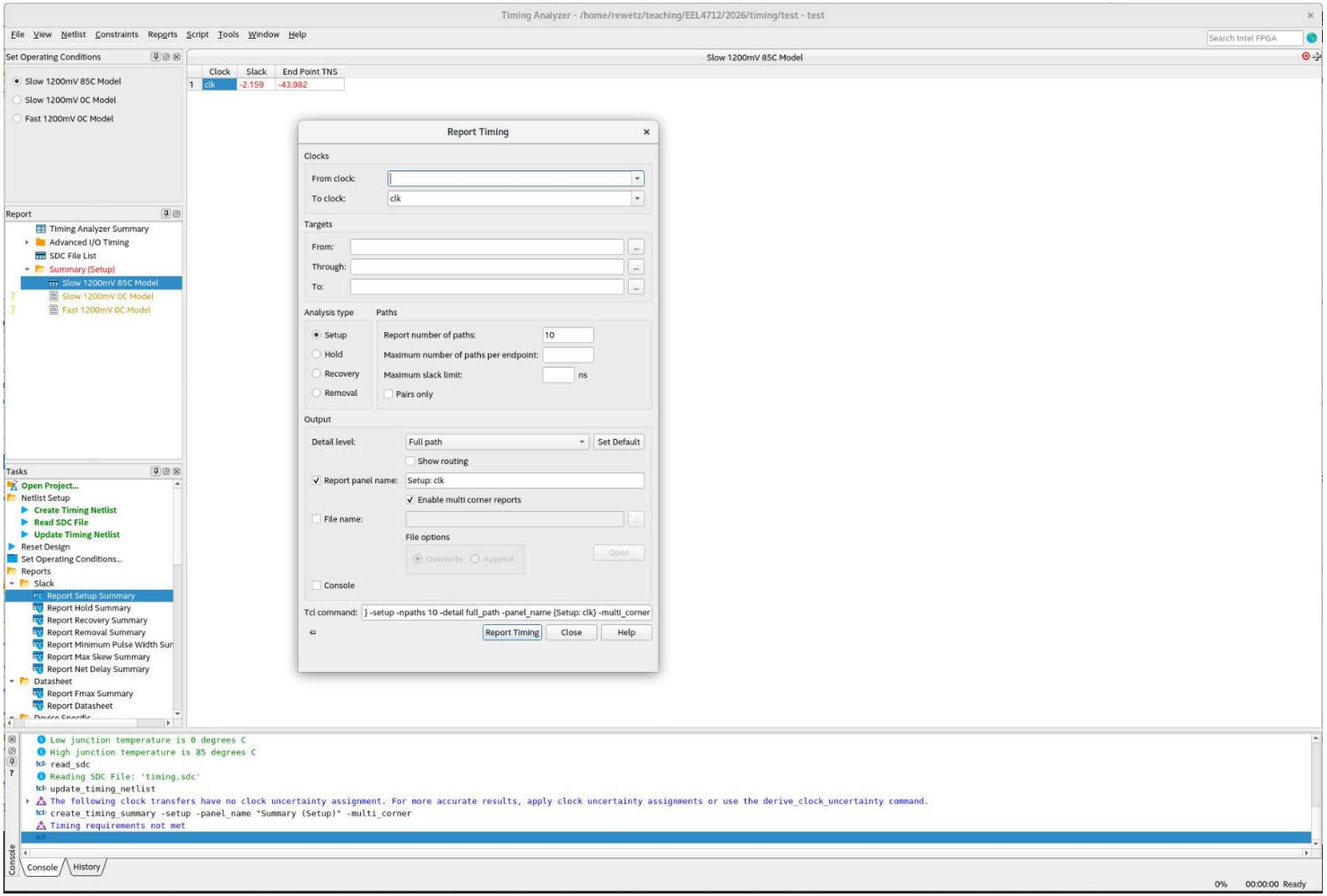
Look at Fmax In Certain Corners



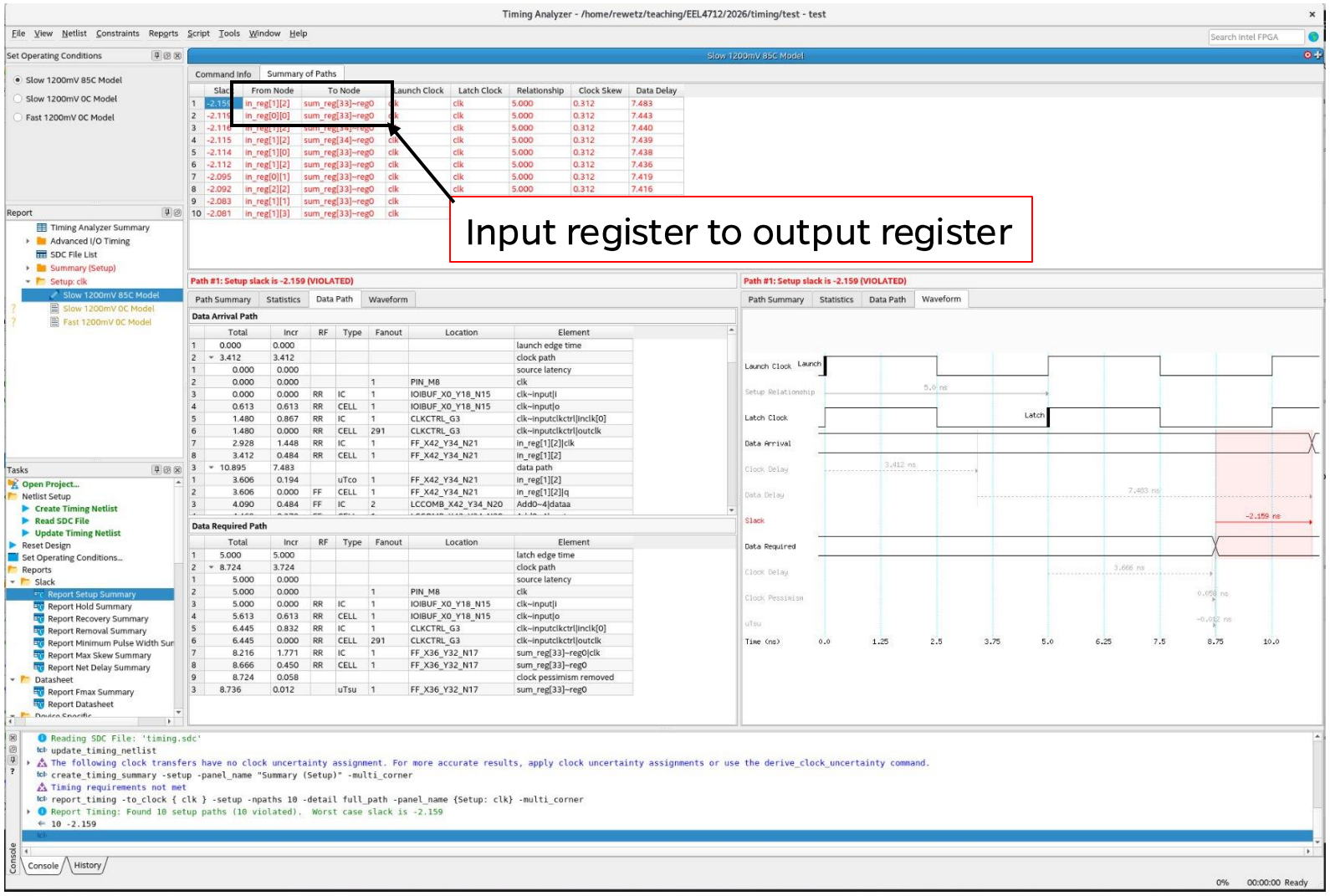
WNS and TNS


Timing Analyzer - /home/rewetz/teaching/EEL4712/2026/timing/test - test
File View Netlist Constraints Reports Script Tools Window Help
Search Intel FPGA
Set Operating Conditions
Slow 1200mV 85C Model
Slow 1200mV 0C Model
Fast 1200mV 0C Model
Slow 1200mV 85C Model
Clock Slack End Point TNS
1 clk -2.159 -43.982
Report
Timing Analyzer Summary
Advanced I/O Timing
SDC File List
Summary (Setup)
Slow 1200mV 85C Model
Slow 1200mV 0C Model
Fast 1200mV 0C Model
Tasks
Open Project...
Netlist Setup
Create Timing Netlist
Read SDC File
Update Timing Netlist
Reset Design
Set Operating Conditions...
Reports
Slack
Report Setup Summary
Report Hold Summary
Report Recovery Summary
Report Removal Summary
Report Minimum Pulse Width Sur
Report Max Skew Summary
Report Net Delay Summary
Datasheet
Report Fmax Summary
Report Datasheet
Low junction temperature is 0 degrees C
High junction temperature is 85 degrees C
read_sdc
Reading SDC File: 'timing.sdc'
update_timing_netlist
The following clock transfers have no clock uncertainty assignment. For more accurate results, apply clock uncertainty assignments or use the derive_clock_uncertainty command.
create_timing_summary -setup -panel_name "Summary (Setup)" -multi_corner
Timing requirements not met
Console
Console History
0% 00:00:00 Ready

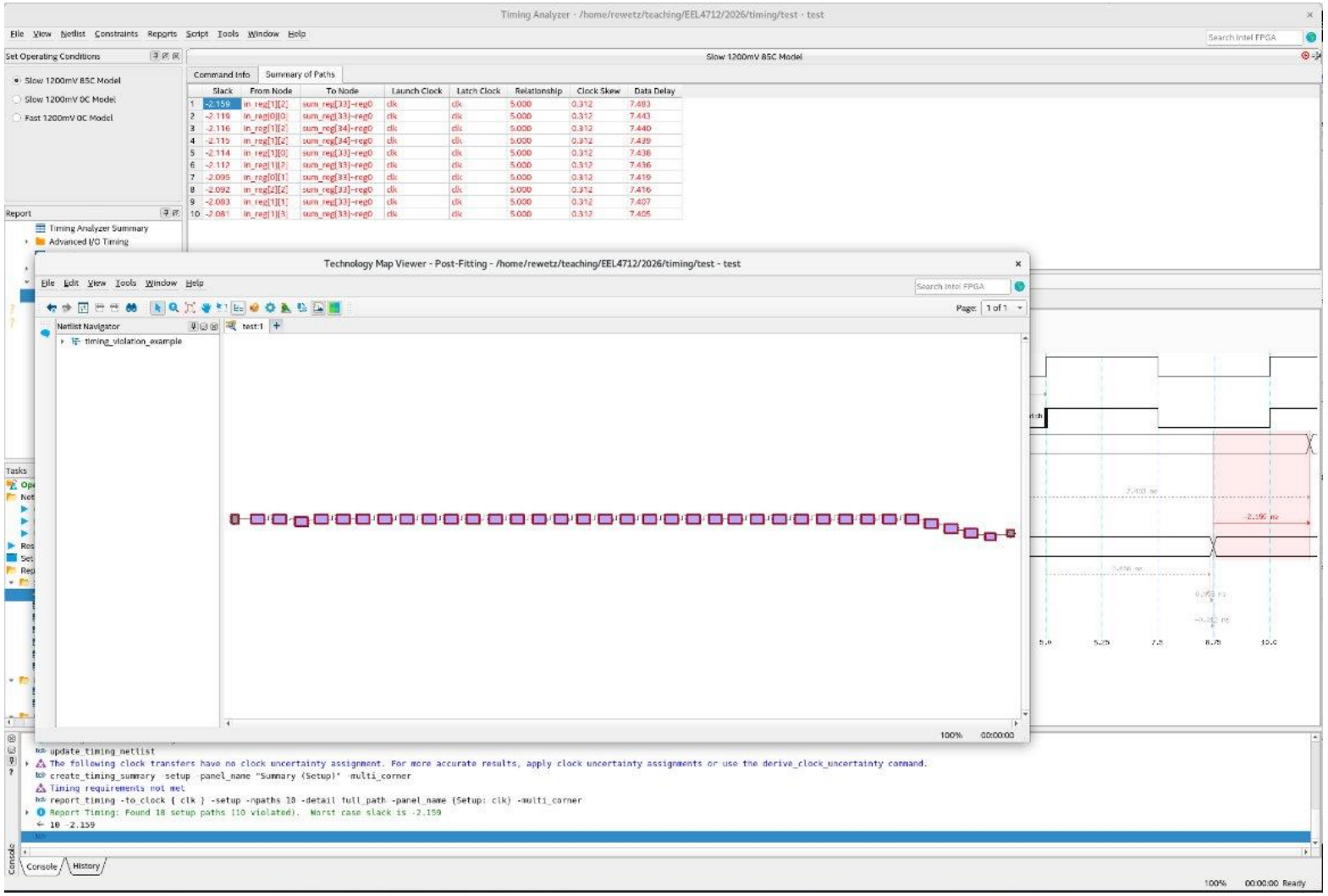
Report Timing using Timing Analyzer



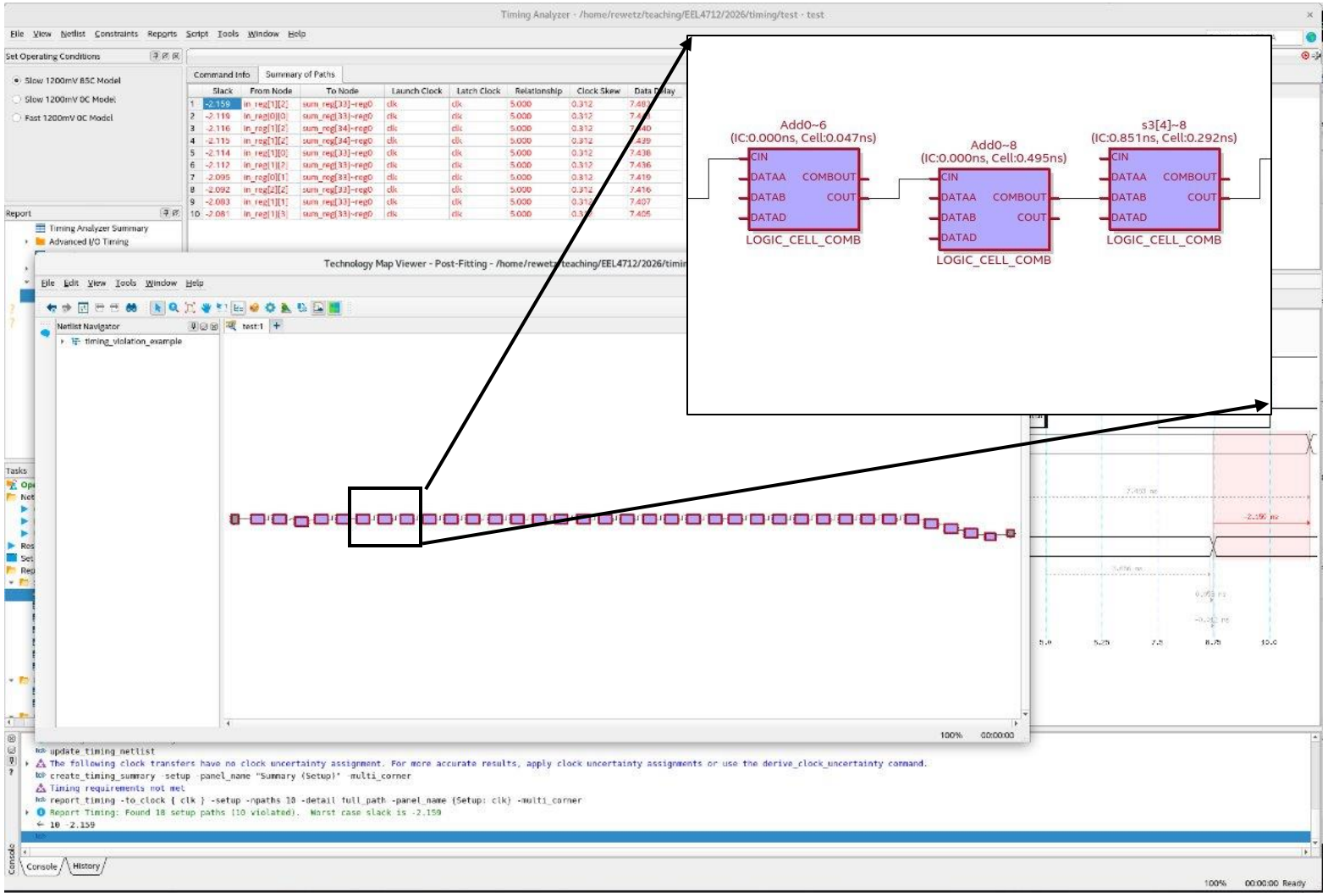
Top 10 Critical Paths



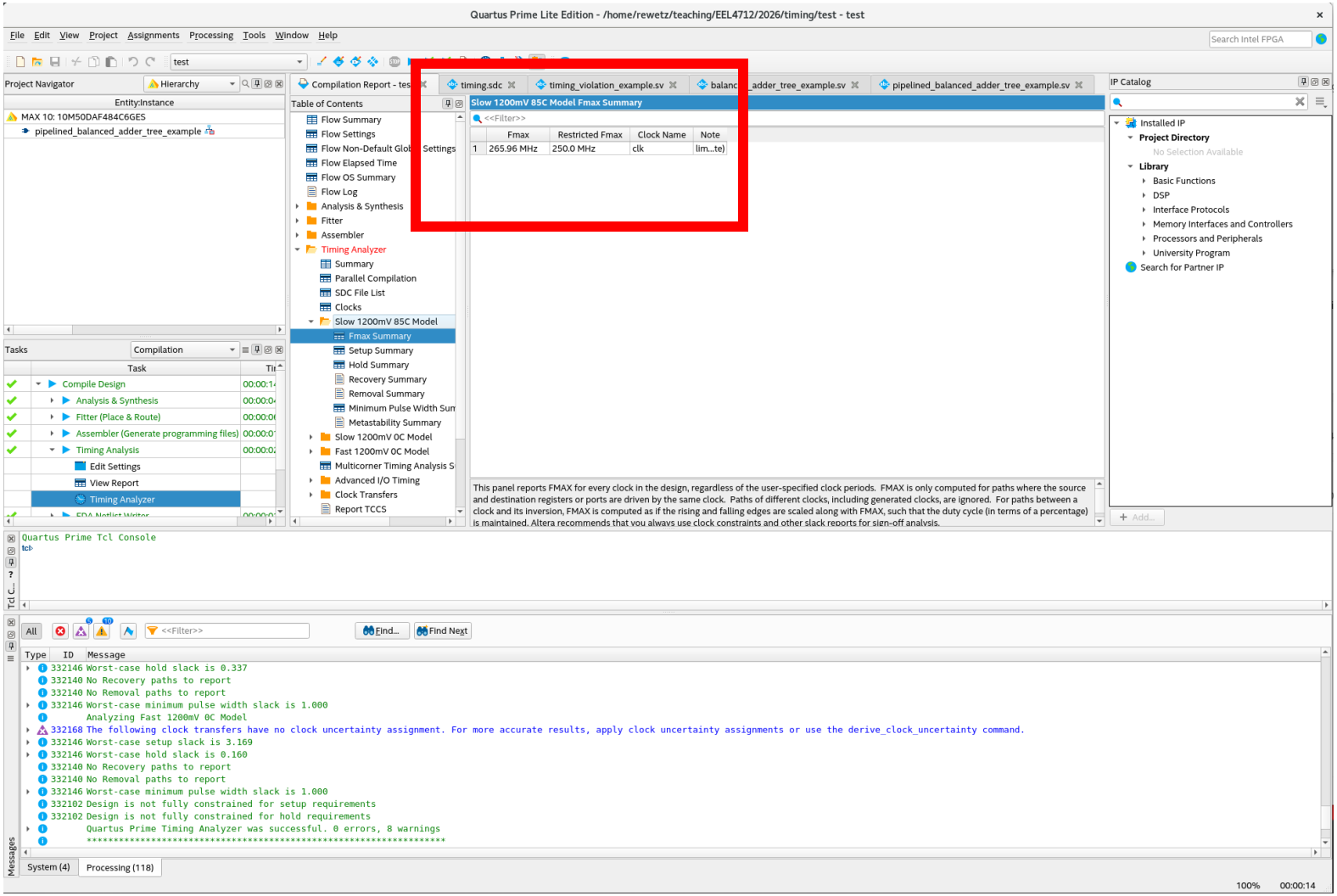
The Critical Path of the Design



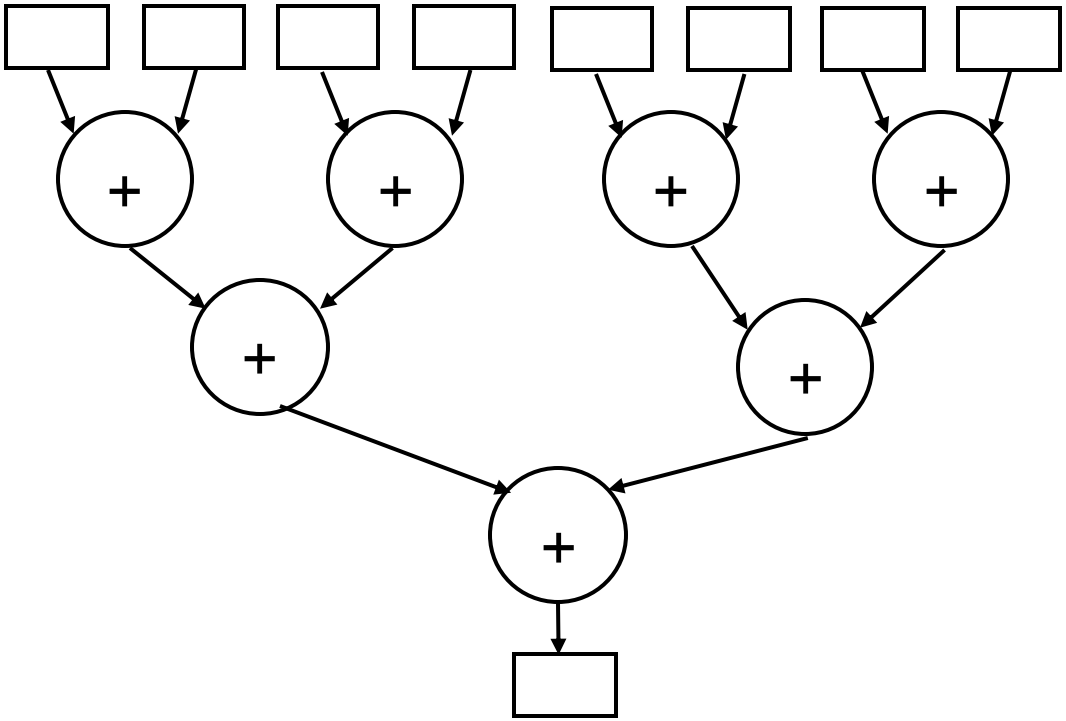
Critical Path Consists of Adders



Fmax > 200 MHz

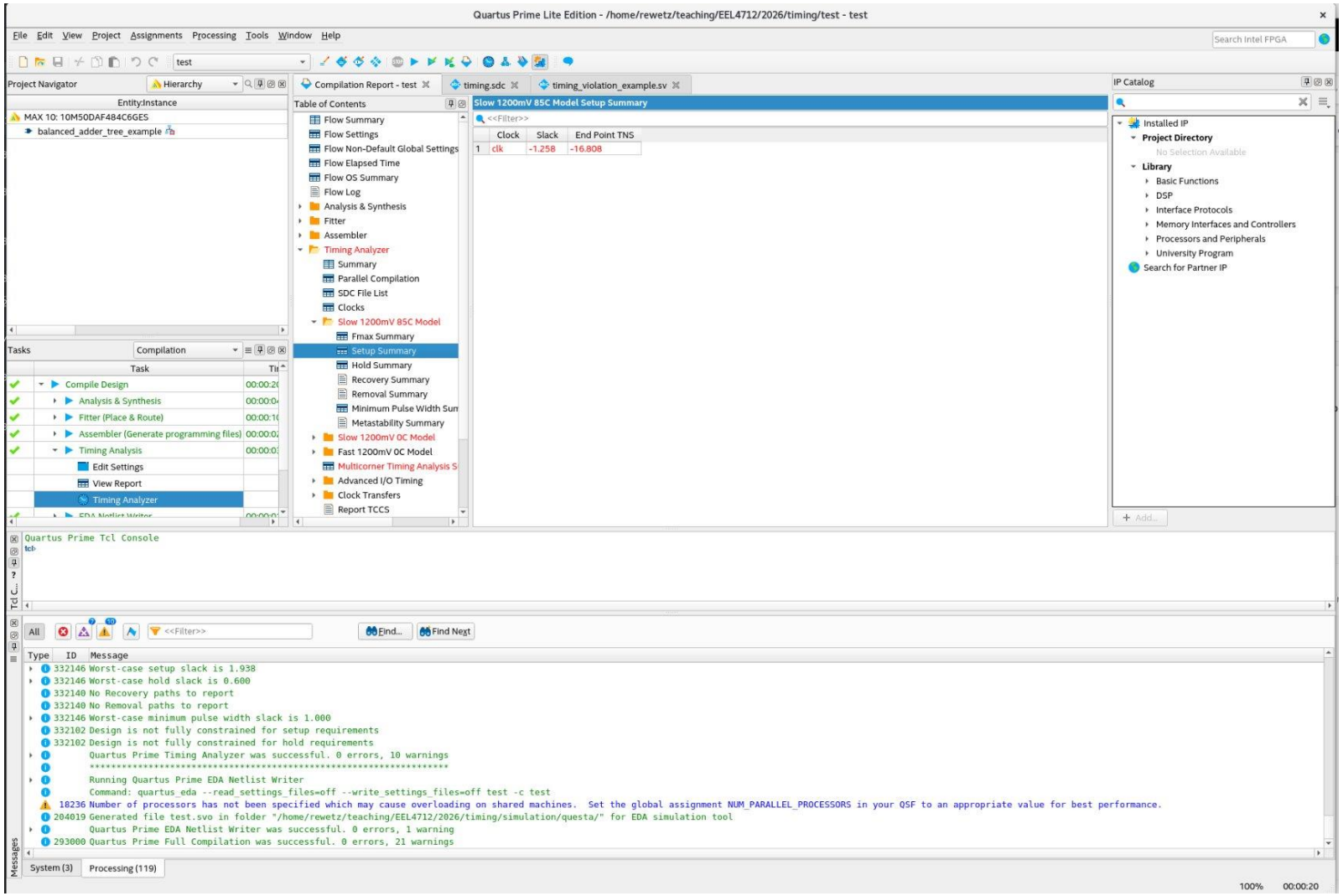

Quartus Prime Lite Edition - /home/rewetz/teaching/EEL4712/2026/timing/test - test
File Edit View Project Assignments Processing Tools Window Help
Search Intel FPGA
Project Navigator
Hierarchy
Entity:Instance
MAX 10: 10M50DAF484C6GES
pipelined_balanced_adder_tree_example
Compilation Report - test
timing.sdc
timing_violation_example.sv
pipelined_balanced_adder_tree_example.sv
Table of Contents
Flow Summary
Flow Settings
Flow Elapsed Time
Flow OS Summary
Flow Log
Analysis & Synthesis
Fitter
Assembler
Timing Analyzer
Summary
Parallel Compilation
SDC File List
Clocks
Slow 1200mV 85C Model
Fmax Summary
Setup Summary
Hold Summary
Recovery Summary
Removal Summary
Minimum Pulse Width Sum
Metastability Summary
Slow 1200mV 0C Model
Fast 1200mV 0C Model
Multicorner Timing Analysis S
Advanced I/O Timing
Clock Transfers
Report TCCS
Slow 1200mV 85C Model Fmax Summary
Fmax
Restricted Fmax
Clock Name
Note
1
265.96 MHz
250.0 MHz
clk
This panel reports FMAX for every clock in the design, regardless of the user-specified clock periods. FMAX is only computed for paths where the source and destination registers or ports are driven by the same clock. Paths of different clocks, including generated clocks, are ignored. For paths between a clock and its inversion, FMAX is computed as if the rising and falling edges are scaled along with FMAX, such that the duty cycle (in terms of a percentage) is maintained. Altera recommends that you always use clock constraints and other slack reports for sign-off analysis.
IP Catalog
Installed IP
Project Directory
Library
Basic Functions
DSP
Interface Protocols
Memory Interfaces and Controllers
Processors and Peripherals
University Program
Search for Partner IP
Tasks
Compilation
Task
Compile Design
Analysis & Synthesis
Fitter (Place & Route)
Assembler (Generate programming files)
Timing Analysis
Edit Settings
View Report
Timing Analyzer
Quartus Prime Tcl Console
Type
ID
Message
332146 Worst-case hold slack is 0.337
332140 No Recovery paths to report
332140 No Removal paths to report
332146 Worst-case minimum pulse width slack is 1.000
Analyzing Fast 1200mV 0C Model
332168 The following clock transfers have no clock uncertainty assignment. For more accurate results, apply clock uncertainty assignments or use the derive_clock_uncertainty command.
332146 Worst-case setup slack is 3.169
332146 Worst-case hold slack is 0.160
332140 No Recovery paths to report
332140 No Removal paths to report
332146 Worst-case minimum pulse width slack is 1.000
332102 Design is not fully constrained for setup requirements
332102 Design is not fully constrained for hold requirements
Quartus Prime Timing Analyzer was successful. 0 errors, 8 warnings
System (4)
Processing (118)
100%
00:00:14

Modify Design To Adder Tree

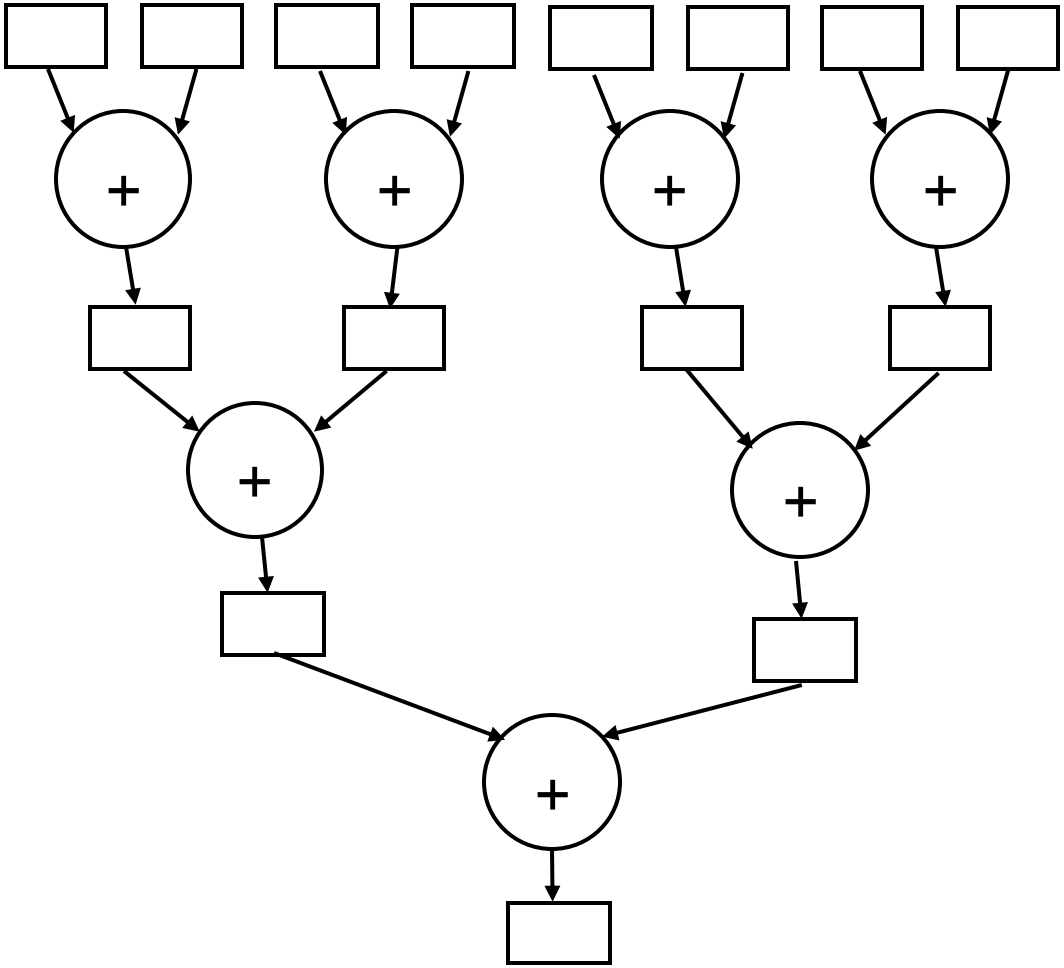


Critical path has only 3 adders!

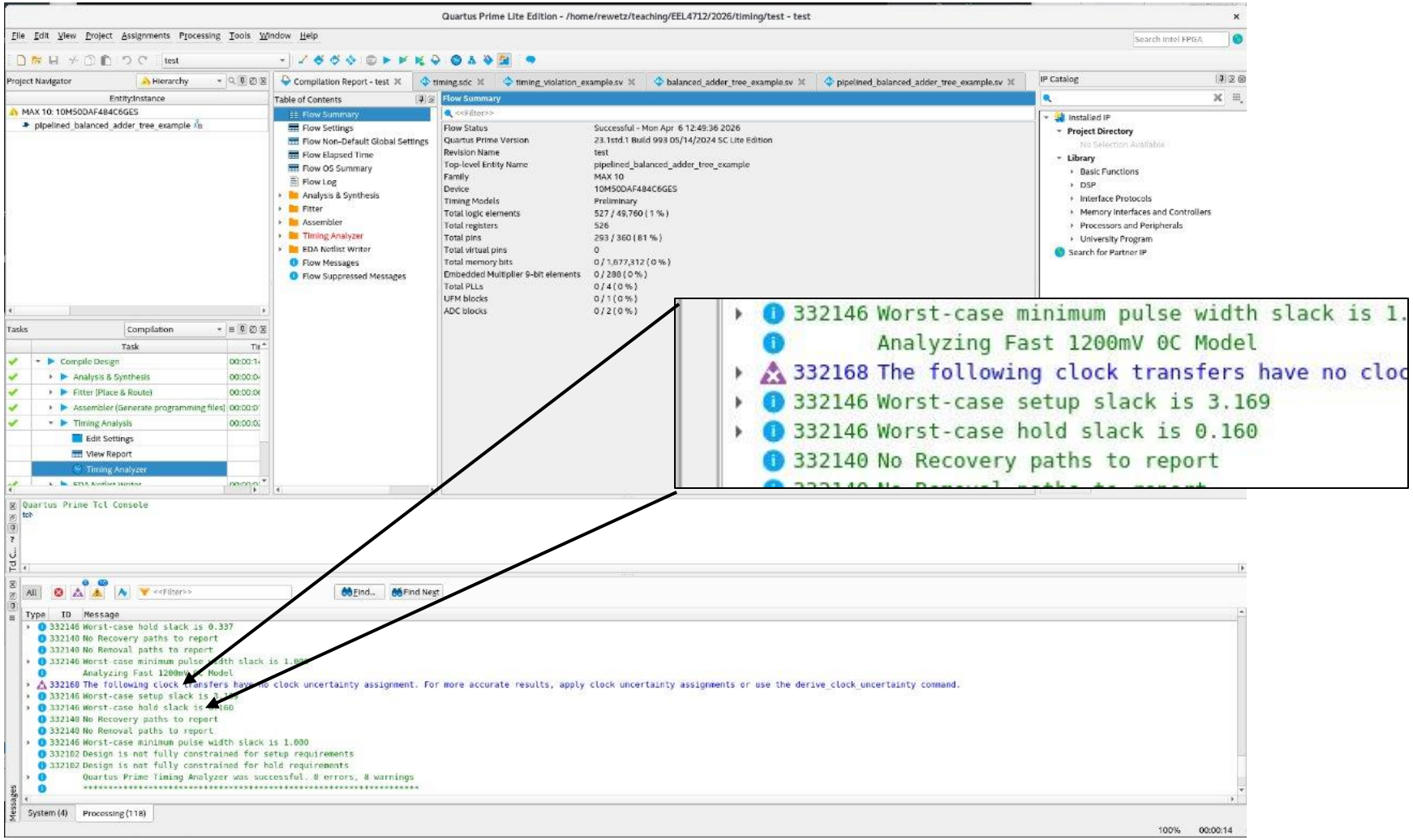
Timing After Change to Adder Tree



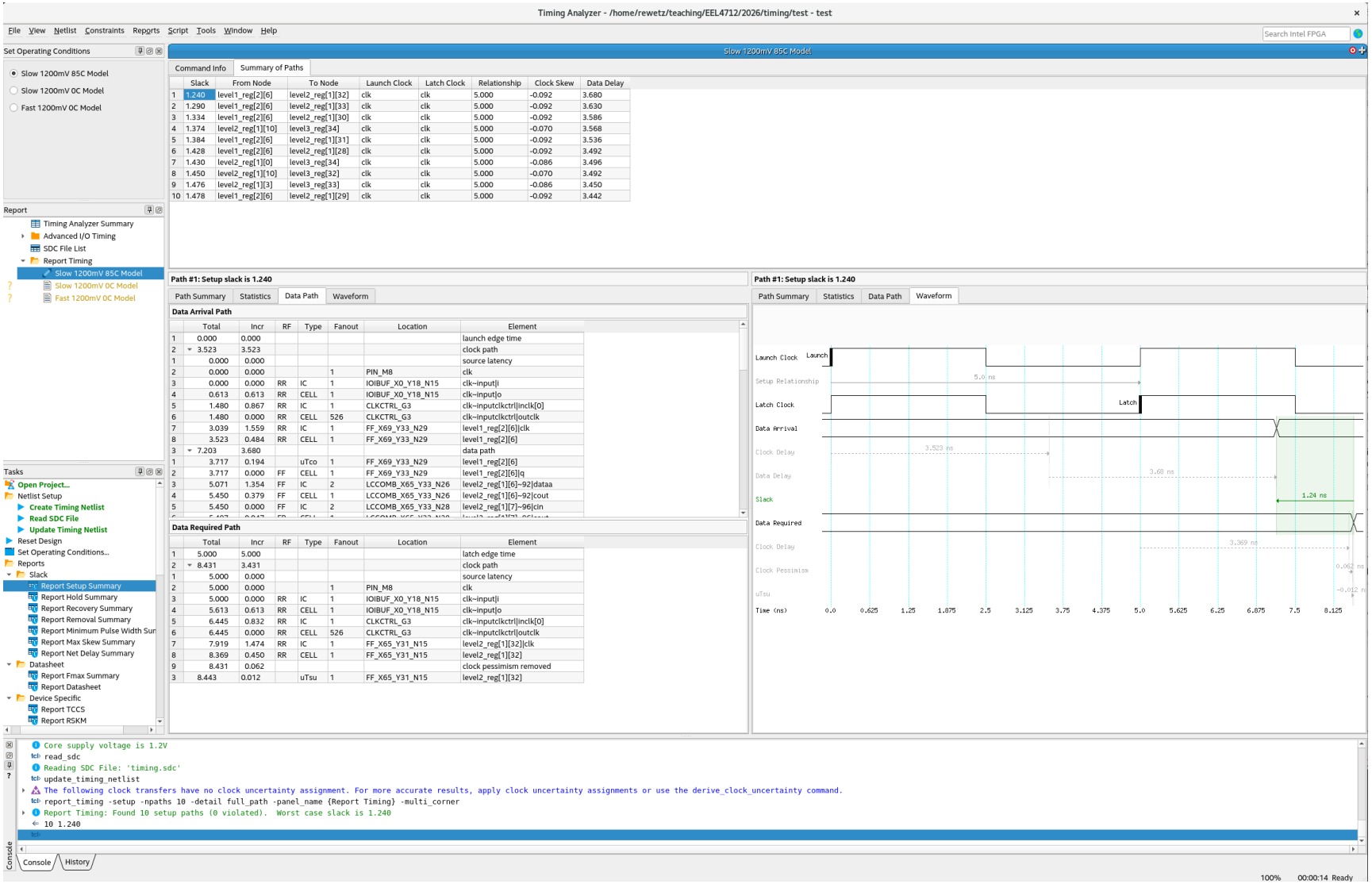
Modify Design To Adder Tree



Pipelined Adder Tree



Positive Slack on Critical Path



Big Picture

- Timing constraints:
  - Define performance target
  - Guide synthesis and placement
  - Ensure hardware works at speed
- If you don’t constrain it:
  - The tool cannot optimize it
  - Silicon will expose the mistake

Summary

- Setup → Maximum delay
- Hold → Minimum delay
- Clock constraints → Required
- I/O constraints → Model the outside world
- False paths → Use carefully
- Multicycle → Only when architecturally intended